

Geometry Outperforms Machine Learning on an Open Cryptic Pocket: Benchmarking Three MD-Free Computational Pocket Predictors on KRAS G12D Switch-II Pocket (SII-P)

¹Ryan Jaemin Ahn

Abstract — Cryptic pockets, protein-binding sites hidden in their ground-state structure and solely formed from flexing, have transformed proteomes traditionally regarded as undruggable into viable targets for drug discovery. Due to the hidden nature of cryptic pockets, researchers increasingly rely on computational tools to predict and locate such pockets. However, the majority of these tools were developed on static, visible protein structures. We benchmarked three molecular-dynamics (MD) free software pocket prediction tools on the KRAS G12D Switch-II Pocket: fpocket, which utilizes Voronoi tessellation and alpha spheres for pocket identification; P2Rank, a machine-learning (ML) trained model for scoring the ligandability of protein surfaces; and AE-PocketMiner, a graph neural network (GNN) trained specifically for prediction of cryptic pocket formation. Tools were run based on two protein structures at the endpoints of the conformational transition and a 24-conformer closed-to-open ladder generated from the normal mode analysis (NMA) of the closed protein structure. On the open structure, fpocket performed the best overall (recall 0.92 and Matthews correlation coefficient 0.95), followed by P2Rank (0.72 and 0.83), with both tools achieving perfect precision. AE-PocketMiner was ranked last overall (0.40 and 0.12), where it flagged 45 residues. On the closed structure, P2Rank alone correctly returned no pockets, while fpocket produced several false positives. Across the ladder, pocket volume was proved to be necessary but insufficient for detection: four conformers at the volume of an open state returned as no pocket at the site, while the neighbouring counterpart was flagged as open. Furthermore, specialization didn't play a role in the detection of the target, and relying on detection rate alone would have concealed AE-PocketMiner's (cryptic pocket specialist) overprediction along with P2Rank's high accuracy in cryptic pocket detection.

I. INTRODUCTION

KRAS is one of the most frequent oncogenic mutations that drive cancer, with pancreatic ductal adenocarcinoma (PDAC) being driven by the G12D mutant in the majority of its cases [1]. KRAS has been deemed undruggable for the past three decades due to its smooth surface that only offers a few pockets deep enough and suitable for a molecule to bind to.

However, Ostrem and colleagues demonstrated that the Switch-II pocket, or SII-P, can actually resolve this dilemma, as a ligand was able to bind to the cavity formed in the KRAS G12C mutant [2], with later studies successfully binding the MRTX1133 inhibitor successfully binding to the G12D mutant counterpart [3,4]. The pocket is cryptic in the sense that its cavities are formed only in a ligand-bound state and remain absent in a ligand-free state. Accounting for these sites is especially imperative for drug discovery, as it significantly increases the number of feasibly druggable protein pockets in disease-associated proteins from 40% to nearly 80% [5,6]. Because cryptic pockets are absent in the proteome's ground state, computational methods used to locate them differ from conventional counterparts.

Three classes of tools are commonly used to computationally predict and locate cryptic pockets. Tools using geometric approaches, such as fpocket, locate pockets in a fixed structure without any training data [7,8]. Trained on databases of known complexes with ligandable neighbouring surfaces, P2Rank utilises machine learning (ML) to serve as a tool for general pocket detection [9,10]. Lastly, there are classes of tools trained specifically to specialize in the prediction of cryptic pocket formation, such as PocketMiner and its attention-aware successor, AE-PocketMiner, which uses graph neural networks (GNN) to predict the probable location where the pocket-opening event would occur [11,12], with related methods typically utilizing AlphaFold to expose the site [13].

Typically, for a cryptic pocket target, the third class of computational tool (AE-PocketMiner) would be the default. Whether this default would be justified when experimenting with a clinically significant cryptic pocket that hasn't been investigated on the open. We therefore asked which of the three classes of computational tools performs the best in reliably localizing the KRAS G12D SII-P across a controlled range of conformational openness and on the experimental protein structures themselves.

II. METHODS

Structures and answer key

Two high-resolution crystal structures of KRAS G12D, one holo and one apo, were obtained from the RCSB Protein Data Bank. Both are GDP-bound, so the only difference between the structures is narrowed down to pocket state. 5US4 is apo with a collapsed SII-P (1.83 Å) in a ligand-free state, and 7RPZ is holo with an open SII-P (1.30 Å) in an MRTX1133 ligand-bound state. Before any of the tools were run, the structures were modified from the removal of the ligand and the solvent. Ground-truth SII-P residues are residues that have atoms within 4.5 Å of MRTX1133 in 7RPZ, which gives 25 residues across Switch-II, helix 3, and the P-loop end of the ligand footprint. Defining the pocket based on what the drug touches, rather than based on any of the tool's output, keeps the answer key independent of the methods that are being assessed. 7RPZ and its use was only limited to defining ground truth, docking validation, and score predictions, and was never used for conformer generation that all derives from the 5US4 structure. Reinforcing this separation matters since a conformer based

¹ Ryan Ahn is with Yongsan International School of Seoul, Seoul, 04347, South Korea (email: 27.ryan.ahn@yiseseoul.org)

on the open structure already contains the answer key, which consequently inflates every downstream result.

Conformational Ladder

Through normal mode analysis via ProDy [14], a 24-conformer ladder was generated from the 5US4 structure. Normal mode analysis assumes the protein behaves like a harmonic oscillating system. By using the Hessian matrix, it breaks down and reanalyzes the protein’s potential energy into collective patterns of atomic oscillation and motion called normal modes. Normal modes exhibiting low frequency correspond to the slow, large-scale movements that often correlate to the opening and closing of protein sites. We specifically displaced a structure along the lowest-frequency mode observed during the opening mode of the switch-II pocket in graded steps, in the absence of molecular dynamics to emulate the sampling regimes that are used in most laboratories. Though molecular dynamics (MD) simulations are well-established techniques in uncovering cryptic pockets, they do require significantly more computation in comparison to MD-free counterparts. Thus, a ladder to MD-free sampling is most appropriate by testing the tools under conditions that will be most pragmatic and useful.

Openness metric and its validation

The openness was quantified as the volume of the pocket located near the His95 Ca recorded by fpocket, and was identified positionally so the same site can be monitored as the protein changes its shape. The metric cleanly separated the anchors: Closed 5US4 with approximately 245 Å³ and a 0.001 druggability score [8] against the open 7RPZ with approximately 822 Å³ and a 0.684 druggability score. Both volume and druggability rise concurrently from a closed-to-open pocket transition between the two anchors, indicating how the metrics track the cryptic pocket formation. Across the ladder, volume rose from approximately 269 Å³ to 750 Å³, which is 91% of the experimental openness volume (Fig. 1). Discontinuity in opening can be noted based on three clusters of volume exhibited rather than in gradients (Table II). Conformer 15 was where the Switch-II wall was gained, which is demonstrated by the immediate rise in the volume from approximately 400 Å³ to 750 Å³. The volume shift thus serves as a core indicator of the moment the actual pocket forms. There are two caveats to note, however. First, since no SII-P exists in the closed state, a small, peripheral cavity is measured and read instead. Second, fpocket is the one measuring in the openness axis while also being one of the tools benchmarked. Though the scoring is unaffected since the answer key itself derives from the drug’s contact rather than from fpocket itself, the axis itself is not independent of the tool.

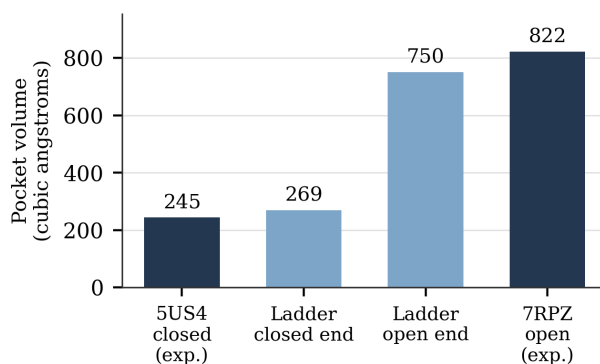


Figure 1. Switch-II pocket volume across closed and open experimental structures and closed and open ends of MD-free conformational ladder. The ladder reaches approximately 91 percent of the open experimental structure.

Docking validation gate

The docking pipeline was validated at the beginning phases of the experiment prior to tool comparison, where MRTX1133 was redocked into a prepared 7RPZ structure through AutoDock Vina [15,16]. The top-ranked pose that was recovered was the crystallographic pose at 0.96 Å, with a predicted affinity of -12.7 kcal/mol.

Pocket prediction tools

The three tools being benchmarked are distinguished by their methodologies in pocket identification. fpocket utilizes Voronoi tessellation and clusters of alpha spheres into different pockets to rank them based on factors including density, hydrophobicity, and polarity, while scoring druggability in separate ways from pocket volume and surface chemistry [7,8]. Furthermore, the tool does so without any training data or understanding of the structure of the binding site. P2Rank utilizes random forest classification that is trained on known complexes to score different solvent-accessible surface points in ligandability while accounting for B-factor, chemical factors, protrusion, and shape features to eventually convert clusters of high-scoring surface points into ranked pockets [9,10]. For 7RPZ, for instance, it identified 2743 individual surface points, with 60 of them being classified as ligandable. Lastly, AE-PocketMiner is distinguished specifically by a graph neural network (GNN) with a residue contact graph trained on pocket openings derived from simulations. Through this, it computes and predicts the residue’s probability of participating in the pocket opening event [11,12]. This characteristic of AE-PocketMiner, emphasizing predicting the location of cryptic pocket opening, contrasts with other computational tools that focus on predicting the location of the pocket the ligand will bind to.

Scoring and metrics

The three tools were all scored based on the same residue sets (chain A of the structure), so the reported metrics are comparable across the tools. fpocket and P2Rank both reported pockets. Specific conditions were applied for reported pockets and residues to be counted, respectively: part of the pocket needed to be within 14 Å of His95, and the residues must be located within 4.5 Å of such a pocket. For instances in which more than one pocket qualified, it

was evaluated via combined residues. 14 Å was set as a boundary to prevent the tools from earning credit by flagging the nucleotide site that is already open in one region of the ligand’s footprint instead of another region with the Switch-II cryptic pocket. We report four metrics for this experiment: recall, precision, distance from predicted pocket to the drug (DCA), and MCC. Of these metrics, MCC is the most important because accounting for every outcome, including what the tools correctly leave alone without flagging, can prevent metric inflation. Due to the AE-PocketMiner’s function being different from the other two tools, a specific cutoff needed to be established to turn probability into a prediction. A standard probability cutoff of 0.7 was used, followed by a validation of the results via comparative analysis of the average probability on true pocket residues versus the average across the entire protein, which is independent of the cutoff.

III. RESULTS

fpocket performed the best on the experimental open structure. Out of 25 ground-truth residues, fpocket was able to identify 23 of them (recall 0.92, MCC 0.95) with a predicted centre (DCA) 2.3 Å from the ligand. P2Rank followed (0.72, 0.83) and DCA 3.6 Å. Both tools achieved perfect precision of 1.00; thus, the difference in the tools’ performance is dependent on coverage rather than false positives (Table 1, Fig. 2). AE-PocketMiner ranked last as it flagged broadly: 45 residues were flagged, indicating it was unable to correctly identify true SII-P residues, with recall 0.40 and MCC 0.12 achieved.

TABLE I. TOOL PERFORMANCE

Tool	Struct.	Recall	MCC	Prec.	Residues
fpocket	7RPZ	0.92	0.95	1.00	23
P2Rank	7RPZ	0.72	0.83	1.00	18
AE-Pocket Miner	7RPZ	0.40	0.12	0.22	45
fpocket	5US4	0.28	0.32	0.54	13
P2Rank	5US4	0.00	0.00	—	0
AE-Pocket Miner	5US4	0.28	0.02	0.16	43

Table 1. Performance of the three tools on two experimental structures (7RPZ: open SII-P, 5US4: closed SII-P) evaluated on recall, MCC, precision, and number of predicted residues.

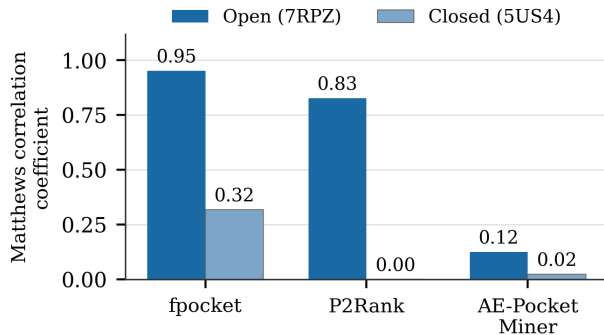


Figure 2. Tool’s localization accuracy on experimental structures. On open (7RPZ), geometric tool fpocket best localizes and AE-PocketMiner ranks last on both. On closed (5US4), P2Rank correctly proposed no pocket at the site, which is a correct negative result.

On the closed structure, P2Rank correctly recalled no pocket within 14 Å of the site. Meanwhile, fpocket exhibited recall 0.28 and precision 0.54, highlighting its high accuracy on open structure being accompanied by false positives on the closed counterpart. AE-PocketMiner again exhibited its behavior of flagging broadly: 43 residues were identified with MCC 0.02.

Mean probability at true SII-P residues was 0.606, contrasting with protein-wide 0.514 on the open structure and 0.537 and 0.514 for true SII-P and protein-wide residues, respectively, on the closed structure (Fig. 3). Such behavior is not a consequence of the 0.7 cutoff nor the generated conformers, since both are experimental structures.

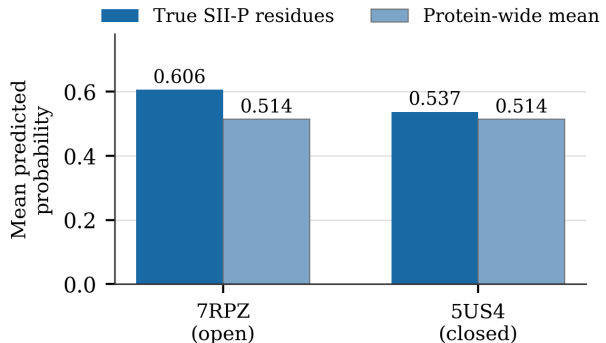


Figure 3. Threshold-independent check on AE-PocketMiner. Mean predicted probability on True SII-P residues against protein-wide one was compared for both open and closed experimental structures. Though noticeable, the difference is smaller for closed.

Across the ladder, the ordering shifts: P2Rank exhibits the highest mean MCC (0.41 ± 0.27), followed by fpocket (0.27 ± 0.07), and AE-PocketMiner is ranked lowest (0.08 ± 0.03) despite achieving the highest mean recall (Fig. 4).

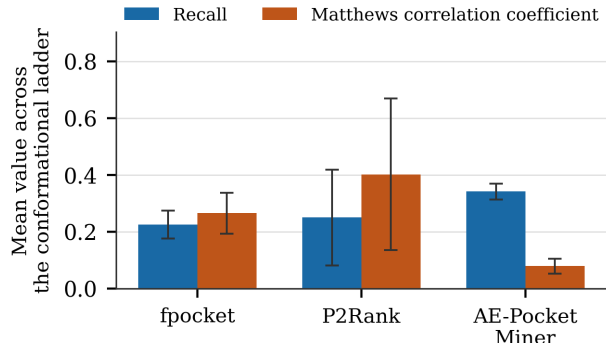


Figure 4. Mean recall and Matthews correlation coefficient (MCC) across the 24-conformer ladder.

P2Rank’s widespread issue has an evident cause: no pockets were found within 14 Å of the site in 7 of 24 conformers, and in four of them at open-state volume, their neighbours of the same volume were detected. Such an

instance reveals how volume is necessary but insufficient, where once the Switch-II wall is present, detection is more dependent on local side-chain arrangement than pocket size.

Where a site is detected, localization and openness are positively correlated: detection rate and recall together at the open end (Table II) when predicted center at a closed cluster to open one moves from 4.4 Å to 3.4-3.9 Å (Fig. 5).

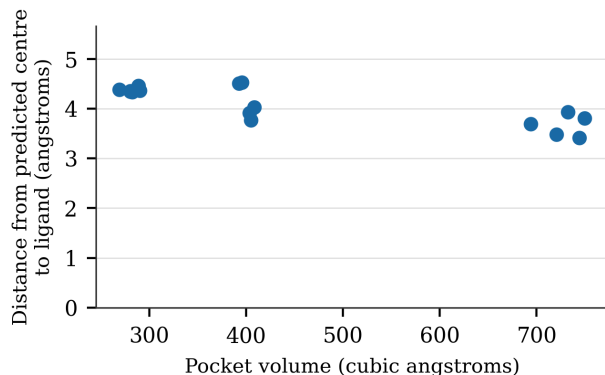


Figure 5. Distance from the predicted pocket centre to the ligand (in angstroms), against pocket volume (in cubic angstroms), for P2Rank across the conformational ladder. Tightening of localization can be observed from about 4.4 Å in closed clusters to 3.4-3.9 Å in open ones.

TABLE II. LADDER AND VOLUME CLUSTER

<i>Volume cluster</i>	<i>n</i>	<i>Range (Å³)</i>	<i>Det.</i>	<i>Mean recall</i>
Closed (~270)	10	269-291	7	0.33
Intermediate (~400)	5	396-421	5	0.33
Open (~720)	9	682-750	5	0.41

Table 2. The ladder is grouped based on volume clusters in three categories: closed, intermediate, and open. Det. counts conformers with a pocket within 14 Å of the site and mean recall is based on those conformers.

IV. DISCUSSION

Two results go against the expectation. First, the purely geometry-based tool outperformed either of the machine learning tools in accurately localizing open cryptic pockets. However, this is unsurprising given that the formed cryptic pocket is geometrically an ordinary concavity, which can then be detected well by the alpha-spheres.

Second, and more importantly, the tool specializing in detecting cryptic pockets ranked last in both experimental structures. Rather than assuming it as a tool defect, we read this as a mismatch between prediction target and task. AE-PocketMiner focuses on estimating a residue's likelihood of participating in pocket opening, which contradicts other tools predicting the location where a ligand would bind. KRAS has multiple regions that are flexible, so there is a structural reason why AE-PocketMiner would flag broadly.

Altogether, the three tools have different positions on the sensitivity-specificity trade-off, and the right choice of which tools to utilize depends on the specific question that is being investigated. P2Rank's correct silence on the closed

pocket, one that reveals the tool's accurate screening workflow in negative control, would be deemed a failure if detection rate is the metric that is being evaluated. fpocket's high accuracy in the open structure comes with the trade-off of the number of false positives it could report when the same tool is tested on a closed structure. Thus, the tool's reliability depends on the task: it can pinpoint the exact boundaries of a structure where the pocket appears open, but it is unreliable for deciding whether the pocket exists in the first place.

The implication of this study is that a tool's stated scope cannot predict how it can perform in a given target class, with the comparison study being cheap to run. The implication for reporting is even sharper. Had detection rate been the only metric reported, the specialist (AE-PocketMiner) would have been called successful for flagging the pocket, with P2Rank as a failure on the closed structure. Both readings are backward, and both are corrected by the localization metric computed from data already in hand.

It's also important to note several limitations that exist. First, conformers derive from NMA alone. The side chains are therefore approximate, and fpocket's fragmentation across the ladder may reflect that instead of the tool itself. Second, fpocket, one of the tools being compared, is the source of the openness axis; though the scoring is unaffected, it does mean the axis isn't independent. Third, KRAS is flexible in multiple regions, meaning AE-PocketMiner's broad flagging can be a behavior specific to this protein rather than a general tendency to overestimate. Lastly, due to the limited scope, and it being one pocket and one target, the failure of the expected ranking with specialists performing the most superior wouldn't be the case in general.

ACKNOWLEDGMENT

We would like to thank Professor Se-Jin Hwang for his guidance and mentorship throughout the research study.

REFERENCES

- [1] A. M. Waters and C. J. Der, "KRAS: The critical driver and therapeutic target for pancreatic cancer," *Cold Spring Harb. Perspect. Med.*, vol. 8, no. 9, a031435, 2018.
- [2] J. M. Ostrem, U. Peters, M. L. Sos, J. A. Wells, and K. M. Shokat, "K-Ras(G12C) inhibitors allosterically control GTP affinity and effector interactions," *Nature*, vol. 503, no. 7477, pp. 548–551, 2013.
- [3] X. Wang et al., "Identification of MRTX1133, a noncovalent, potent, and selective KRAS G12D inhibitor," *J. Med. Chem.*, vol. 65, no. 4, pp. 3123–3133, 2022.
- [4] J. Hallin et al., "Anti-tumor efficacy of a potent and selective non-covalent KRAS G12D inhibitor," *Nat. Med.*, vol. 28, no. 10, pp. 2171–2182, 2022.
- [5] P. Cimermancic et al., "CryptoSite: Expanding the druggable proteome by characterization and prediction of cryptic binding sites," *J. Mol. Biol.*, vol. 428, no. 4, pp. 709–719, 2016.
- [6] V. Oleinikovas, G. Saladino, B. P. Cossins, and F. L. Gervasio, "Understanding cryptic pocket formation in protein targets by enhanced sampling simulations," *J. Am. Chem. Soc.*, vol. 138, no. 43, pp. 14257–14263, 2016.
- [7] V. Le Guilloux, P. Schmidtke, and P. Tuffery, "Fpocket: An open source platform for ligand pocket detection," *BMC Bioinformatics*, vol. 10, no. 168, 2009.

- [8] P. Schmidtke and X. Barril, "Understanding and predicting druggability: A high-throughput method for detection of drug binding sites," *J. Med. Chem.*, vol. 53, no. 15, pp. 5858–5867, 2010.
- [9] R. Krivák and D. Hoksza, "P2Rank: Machine learning based tool for rapid and accurate prediction of ligand binding sites from protein structure," *J. Cheminform.*, vol. 10, no. 39, 2018.
- [10] L. Jendele, R. Krivak, P. Skoda, M. Novotny, and D. Hoksza, "PrankWeb: A web server for ligand binding site prediction and visualization," *Nucleic Acids Res.*, vol. 47, no. W1, pp. W345–W349, 2019.
- [11] A. Meller et al., "Predicting locations of cryptic pockets from single protein structures using the PocketMiner graph neural network," *Nat. Commun.*, vol. 14, no. 1177, 2023.
- [12] S. Zhang, P. Mishra, D. Kelly, R. Kumar, and G. R. Bowman, "AE-PocketMiner uses attention to simultaneously predict cryptic pockets and their allosteric coupling," *bioRxiv*, 2026, doi: 10.64898/2026.05.21.726899.
- [13] A. Meller, S. Bhakat, S. Solieva, and G. R. Bowman, "Accelerating cryptic pocket discovery using AlphaFold," *J. Chem. Theory Comput.*, vol. 19, no. 14, pp. 4355–4363, 2023.
- [14] A. Bakan, L. M. Meireles, and I. Bahar, "ProDy: Protein dynamics inferred from theory and experiments," *Bioinformatics*, vol. 27, no. 11, pp. 1575–1577, 2011.
- [15] O. Trott and A. J. Olson, "AutoDock Vina: Improving the speed and accuracy of docking with a new scoring function, efficient optimization, and multithreading," *J. Comput. Chem.*, vol. 31, no. 2, pp. 455–461, 2010.
- [16] J. Eberhardt, D. Santos-Martins, A. F. Tillack, and S. Forli, "AutoDock Vina 1.2.0: New docking methods, expanded force field, and Python bindings," *J. Chem. Inf. Model.*, vol. 61, no. 8, pp. 3891–3898, 2021.
- [17] D. Chicco and G. Jurman, "The advantages of the Matthews correlation coefficient (MCC) over F1 score and accuracy in binary classification evaluation," *BMC Genomics*, vol. 21, no. 6, 2020.